BITS PILANI WILP · AIMLZG529: DATA MANAGEMENT FOR MACHINE LEARNING · ASSIGNMENT 2

End-to-End Data Management Pipeline for a Recommendation System

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GitHub: https://github.com/chandan2738/movielens-recsys-data-pipeline

100,000	26 Features	+111.5%	4.497s
Validated Interactions (943 Users · 1,682 Movies)	Unified Feature Store (Train & Inference)	NDCG@10 Lift (0.0792 → 0.1675)	End-to-End Prefect 3 Orchestration Run

1. Objective & Dataset Overview

This project implements a modular, reproducible, end-to-end recommendation data management pipeline built over the public **MovieLens 100K** dataset (`ml-100k` from GroupLens), comprising `943` users, `1,682` movies across 19 genre flags, and `100,000` explicit 1–5 ratings recorded between `1997-09-20` and `1998-04-22`. To verify real-world data governance, Stage 1 ingestion appends `17` documented malformed audit rows (missing primary keys, invalid types, out-of-range ratings/timestamps, duplicates, and foreign-key violations) that Stage 2 automatically catches and isolates into `data/errors/`.

2. Pipeline Architecture

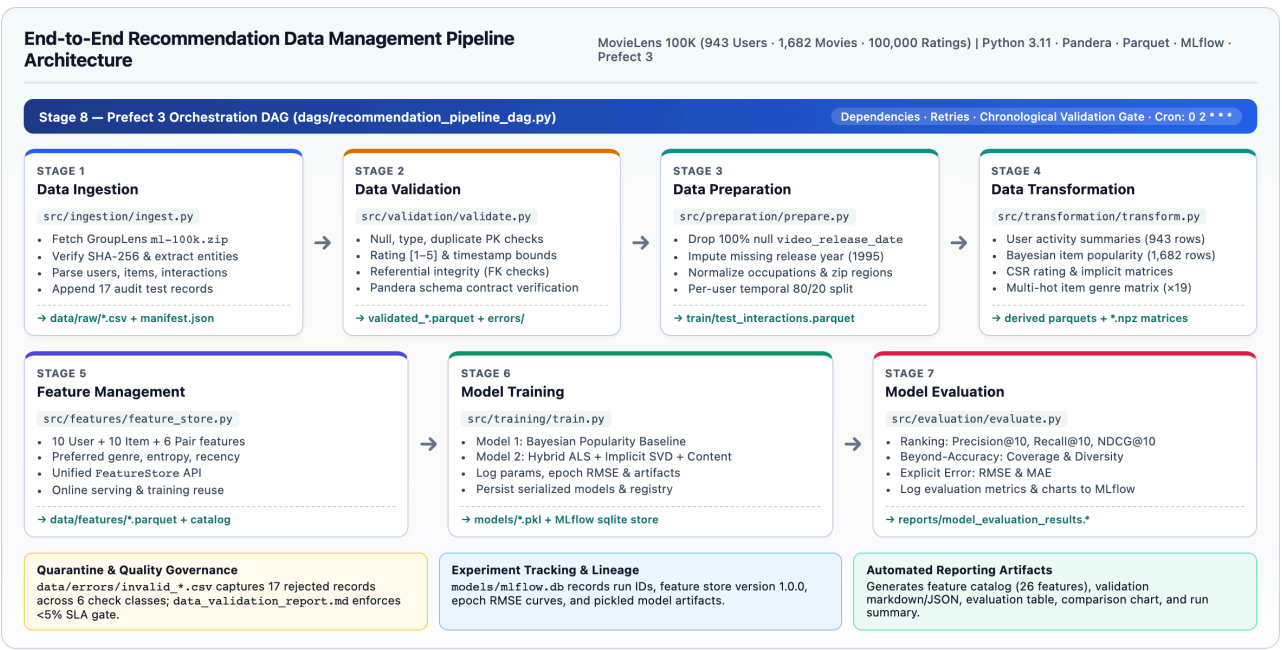


Figure 1: Eight-stage modular architecture orchestrated by Prefect 3 with Pandera validation gates, Parquet feature tables, and MLflow experiment tracking.

3. Tools Selected & Engineering Design Decisions

Tool Stack

- **Data & Storage:** Python 3.11, Pandas, PyArrow (Parquet), SciPy CSR matrices (`94.98%` sparse representation).
- **Data Quality:** `Pandera` declarative schema contracts paired with custom row-level quarantine writers.
- **Feature Store:** Custom Parquet-backed `FeatureStore` serving both batch training and online candidate scoring.
- **Modeling & Tracking:** NumPy Biased ALS + Scikit-learn `TruncatedSVD` + Content Cosine Affinity, tracked in `MLflow 3.18` (SQLite backend).
- **Orchestration:** `Prefect 3` DAG (`@flow` + 7 `@task` nodes) with retries, timing telemetry, and cron schedule `0 2 * * *`.

Key Design Rationale

- **Per-User Temporal 80/20 Split:** MovieLens 100K users rated movies in short bursts; a naive global cutoff at 80% leaves only `109` of `943` users overlapping train and test. Sorting chronologically per user ensures zero future leakage while preserving `943/943` user overlap (`79,619` train / `20,381` test).
- **Strict Leakage Prevention:** Derived summaries, interaction matrices (`943×1682`), and all 26 features are computed exclusively on `train_interactions.parquet`.
- **Hybrid Blending:** Explicit Biased ALS minimizes rating error (RMSE/MAE), Implicit SVD captures positive co-watch behavior (`rating ≥ 4.0`), and Feature Store genre cosine similarity mitigates niche/cold-item gaps.

4. Data Validation & Quarantine Governance (Stage 2)

Stage 2 enforces six automated validation rules before any record enters downstream preparation: missing required fields, integer/float type coercibility, primary-key and interaction uniqueness, domain bounds (age `[5, 110]`, gender `{M, F}`, rating `[1.0, 5.0]`), timestamp bounds (`1997-01-01` to `2030-01-01`), and foreign-key referential integrity against validated user and item masters. Clean records are verified with lazy **Pandera** schema contracts and must satisfy the `<5.0%` error-rate SLA gate.

Dataset	Raw Staged Rows	Clean Validated Rows	Quarantined Rows	Error Rate	Quarantined Anomaly Classes	Pandera Contract
`users`	947	943	4	0.422%	Null PK (1), Invalid Type (1), Domain Violation (1), Duplicate PK (1)	**PASSED**
`items`	1,685	1,682	3	0.178%	Empty Title (1), Non-Int ID (1), Duplicate PK (1)	**PASSED**
`interactions`	100,010	100,000	10	0.010%	Missing Fields (2), Type Error (1), Rating Bounds (2), Bad Timestamp (2), FK Orphan (2), Duplicate (1)	**PASSED**

> **Source Null Audit & Preparation Actions:** In Stage 3 Preparation, the 100%-empty `video_release_date` column (`1,682` nulls) was dropped, `1` missing `release_date` was imputed to catalog median year `1995`, `3` missing IMDb URLs were normalized to empty strings, and zip codes were mapped to regional buckets.

5. Feature Management & Feature Catalog (Stage 5)

`src/features/feature_store.py` materializes **26 documented features** across User (10), Item (10), and Interaction (6) entities into Parquet tables and exposes `FeatureStore.build_candidate_features(user_id, candidate_item_ids)` so online candidate scoring reuses identical feature definitions and metadata (`feature_store_version = 1.0.0`).

#	Feature Name	Entity	Dtype	Computation / Definition	Source	Reuse Scope
1	`user_avg_rating`	USER	`float64`	Mean historical rating given by user on train split	`train_interactions`	Training & Inference
2	`user_rating_std`	USER	`float64`	Standard deviation of user's training ratings	`train_interactions`	Training & Inference
3	`user_activity_count`	USER	`int64`	Total number of interactions in user's train history	`train_interactions`	Training & Inference
4	`user_positive_rate`	USER	`float64`	Share of user ratings >= 4.0	`train_interactions`	Training & Inference
5	`user_preferred_genre`	USER	`string`	Top genre weighted by user ratings across rated items	`train_interactions + items`	Training & Inference
6	`user_genre_entropy`	USER	`float64`	Shannon entropy over user's rated genre distribution	`train_interactions + items`	Training & Inference
7	`user_days_since_last_interaction`	USER	`float64`	Days from user's last train interaction to train cutoff	`train_interactions`	Training & Inference
8	`user_age`	USER	`int64`	Validated user age in years	`users_clean`	Training & Inference
9	`user_gender_code`	USER	`int64`	Binary encoding of gender (M=1, F=0)	`users_clean`	Training & Inference
10	`user_occupation_code`	USER	`int64`	Deterministic integer encoding of occupation category	`users_clean`	Training & Inference
11	`item_popularity_count`	ITEM	`int64`	Total training rating count received by item	`train_interactions`	Training & Inference
12	`item_popularity_log`	ITEM	`float64`	Log-scaled popularity log(1 + item_popularity_count)	`train_interactions`	Training & Inference
13	`item_avg_rating`	ITEM	`float64`	Mean training rating received by item (prior fallback)	`train_interactions`	Training & Inference
14	`item_bayesian_rating`	ITEM	`float64`	Bayesian smoothed score (v*R + m*C)/(v+m)	`train_interactions`	Training & Inference
15	`item_positive_rate`	ITEM	`float64`	Fraction of item ratings >= 4.0	`train_interactions`	Training & Inference
16	`item_release_year`	ITEM	`int64`	Cleaned movie release year	`items_clean`	Training & Inference
17	`item_age_years`	ITEM	`int64`	Movie age in years relative to dataset reference year (1998)	`items_clean`	Training & Inference
18	`item_primary_genre`	ITEM	`string`	Primary genre label assigned to item	`items_clean`	Training & Inference
19	`item_genre_count`	ITEM	`int64`	Number of genre tags attached to movie	`items_clean`	Training & Inference
20	`item_days_since_last_interaction`	ITEM	`float64`	Days elapsed since item's most recent train rating	`train_interactions`	Training & Inference
21	`user_item_bias_baseline`	INTERACTION	`float64`	Additive baseline global_mean + b_u + b_i clipped to [1, 5]	`user + item features`	Training & Inference
22	`user_preferred_genre_match`	INTERACTION	`float64`	Cosine affinity between user genre profile and item genres	`user_genre_profile + items`	Training & Inference
23	`is_preferred_genre`	INTERACTION	`int64`	Indicator (1/0) whether item includes user's top genre	`user + item features`	Training & Inference
24	`interaction_recency_days`	INTERACTION	`float64`	Days elapsed between interaction timestamp and train cutoff	`train_interactions`	Training Only
25	`is_weekend_interaction`	INTERACTION	`int64`	Indicator (1/0) whether interaction occurred on Sat/Sun	`train_interactions`	Training & Context
26	`rating_deviation_from_user_mean`	INTERACTION	`float64`	Observed rating minus user_avg_rating	`train_interactions`	Training Target Residual

6. Model Training, Experiment Tracking & Evaluation (Stages 6–7)

Two distinct recommendation models were trained exclusively on the `79,619` training interactions and tracked in **MLflow** (experiment `movielens-100k-recommender`, `sqlite:///models/mlflow.db`): (1) **Baseline Popularity** ranking unseen items via Bayesian smoothed ratings (`m = 25`) plus log-popularity bonus (trained in `0.018s`), and (2) **Hybrid Personalized SVD** combining 16-factor Biased ALS Matrix Factorization (`20` epochs, `reg = 18.0`, final train RMSE `0.8194`), 24-factor Implicit `TruncatedSVD` on L2-normalized positive preferences (`rating ≥ 4.0`), and Feature Store genre cosine affinity (weights `0.15 / 0.75 / 0.10`, trained in `1.139s`).

Evaluation Metric (K = 10, Relevant ≥ 4.0)	Baseline Popularity	Hybrid Personalized SVD	Relative Improvement	Interpretation
Precision@10 (higher is better)	0.0650	**0.1239**	**+90.6%**	Nearly 2× higher relevant hit rate in top-10 lists
Recall@10 (higher is better)	0.0554	**0.1360**	**+145.5%**	Captures 13.6% of held-out user favorites in first 10 slots
NDCG@10 (higher is better)	0.0792	**0.1675**	**+111.5%**	Strong rank-position ordering quality across all 943 users
Catalog Coverage@10 (higher is better)	0.0392	**0.1718**	**+338.3%**	Surfaces 289 distinct titles vs. 66 blockbuster-only items
Intra-List Diversity@10 (higher is better)	0.7127	**0.5940**	-16.7%	Expected focus shift toward user's preferred genres
RMSE (lower is better)	1.0991	**0.9703**	**+11.7%**	Reduced explicit rating squared error across 20,381 test ratings
MAE (lower is better)	0.8870	**0.7690**	**+13.3%**	Sub-0.77 star mean prediction error on held-out ratings

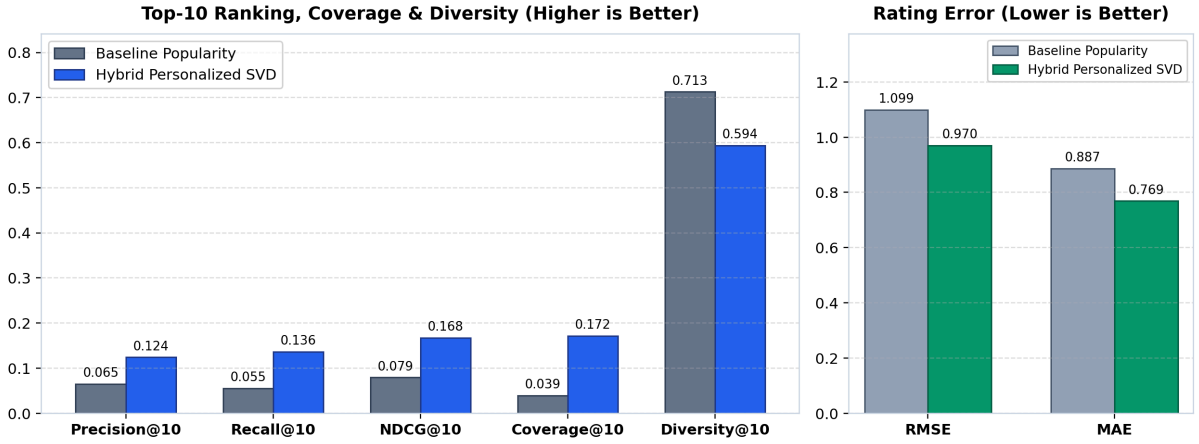


Figure 2: Side-by-side comparison of Top-10 ranking, coverage, diversity, and explicit rating error across all 943 users.

> **Qualitative Personalization Check (`user_id = 196`, preferred genre = `Comedy`):** Top-5 Hybrid recommendations returned *Monty Python and the Holy Grail (1974)* [Comedy, 0.4606], *When Harry Met Sally... (1989)* [Comedy|Romance, 0.4530], *Young Frankenstein (1974)* [Comedy|Horror, 0.4154], *Monty Python's Life of Brian (1979)* [Comedy, 0.4015], and *Annie Hall (1977)* [Comedy|Romance, 0.3959].

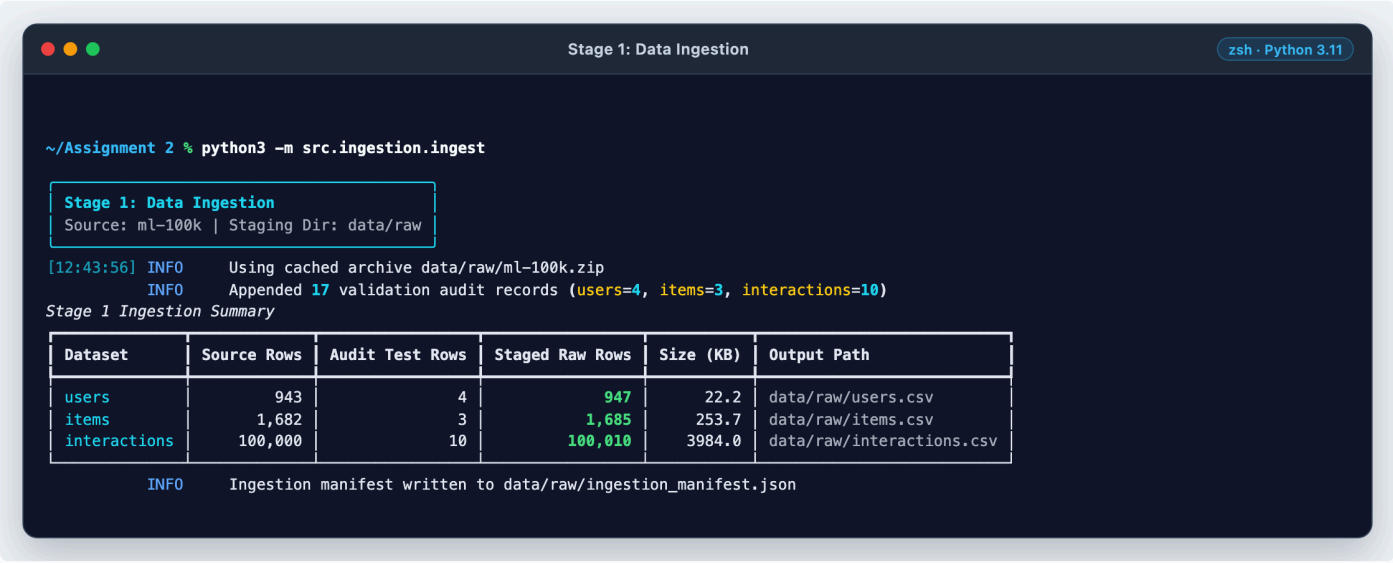
7. Limitations & Future Work

- **Cold-Start Entities:** Brand-new users without interactions fall back to Bayesian popularity and demographic/genre defaults; adding an item/user metadata neural tower (e.g., LightFM or Two-Tower network) would strengthen cold-start personalization.
- **Accuracy–Diversity Trade-off:** Genre affinity weighting raised relevance by >90% while lowering intra-list diversity from `0.7127` to `0.5940`. Applying Maximal Marginal Relevance (MMR) post-ranking can re-balance genre exploration.
- **Single-Node Execution Scale:** Parquet, SQLite MLflow, and embedded Prefect execute the entire 100K workflow in `4.497s` locally. Scaling to 100M+ events requires migrating storage to Apache Iceberg/PostgreSQL, Feast online Redis store, and remote Prefect workers.

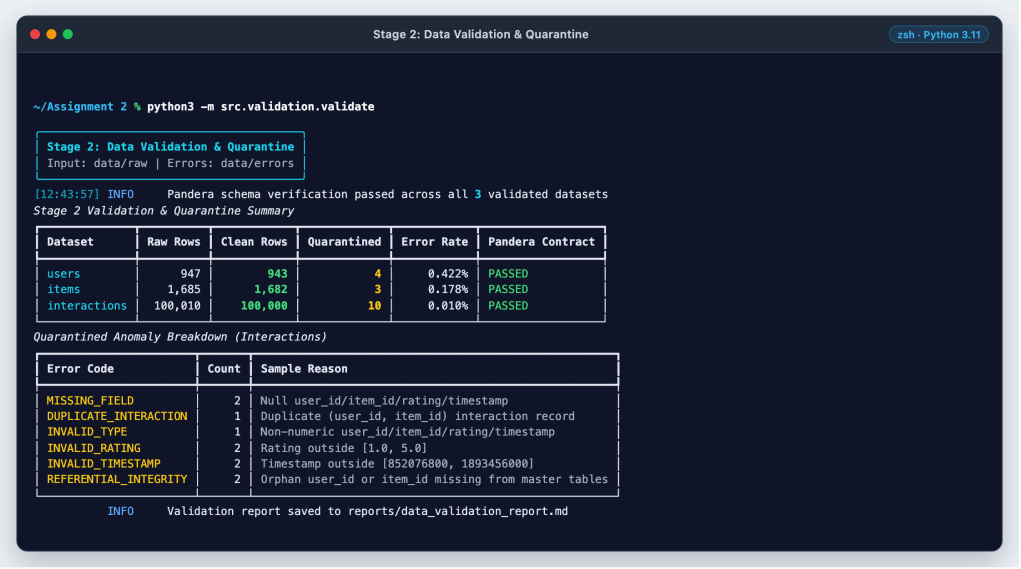
Appendix A: Pipeline Stage Execution Screenshots (Stages 1–3)

GitHub: https://github.com/chandan2738/movielens-recsys-data-pipeline

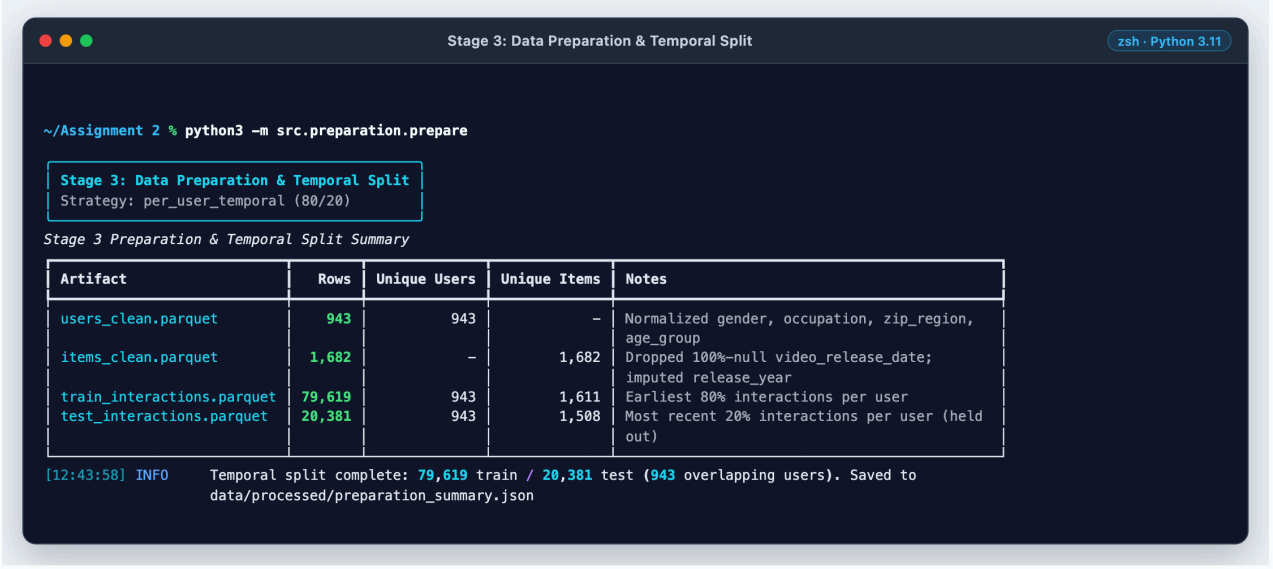
Stage 1 — Data Ingestion (`python3 -m src.ingestion.ingest`)



Stage 2 — Automated Data Validation, Pandera Contracts & Quarantine (`python3 -m src.validation.validate`)



Stage 3 — Data Preparation & Per-User Temporal 80/20 Split (`python3 -m src.preparation.prepare`)



Stage 4 — Data Transformation & Sparse CSR Interaction Matrices (`python3 -m src.transformation.transform`)

Stage 4: Data Transformation & Matrices — zsh · Python 3.11

```
~/Assignment 2 % python3 -m src.transformation.transform

┌──────────────────────────────────────────────────────────┐
│ Stage 4: Data Transformation & Derived Datasets          │
│ Source: data/processed (leakage-free train split)        │
└──────────────────────────────────────────────────────────┘
Stage 4 Derived Datasets & Interaction Matrices
```

Derived Artifact	Shape / Rows	Details
user_activity_summary.parquet	943 rows	Avg rating, activity count, span days, genre breadth
item_popularity.parquet	1,682 rows	Popularity rank, Bayesian weighted rating, positive rate
user_item_rating_matrix.npz	943 x 1682	CSR float32 | NNZ=79,619 | Sparsity=94.98%
user_item_implicit_matrix.npz	943 x 1682	Positive ratings >= 4.0 | Positives=45,602
item_genre_matrix.npz	1682 x 19	Multi-hot item genre matrix aligned to item_idx

```
[12:43:59] INFO     Derived datasets saved to data/processed
```

Stage 5 — Feature Store Materialization & Online Serving Verification (`python3 -m src.features.feature_store`)

Stage 5: Feature Management & Online Reuse — zsh · Python 3.11

```
~/Assignment 2 % python3 -m src.features.feature_store

┌──────────────────────────────────────────────────────────┐
│ Stage 5: Feature Management & Feature Store               │
│ Output Store: data/features | Catalog: reports/feature_catalog.md │
└──────────────────────────────────────────────────────────┘
Stage 5 Feature Store Materialization
```

Feature Table	Rows	Cols	Artifact File
user_features	943	10	user_features.parquet
item_features	1,682	10	item_features.parquet
interaction_features	79,619	6	interaction_features.parquet

Online Inference Feature Serving Check (user_id=196)

Item	UserPref	ItemGenre	Bayes	BiasPred	GenreSim	PrefHit
1	Comedy	Animation	3.862	3.944	0.5268	1
50	Comedy	Action	4.317	4.416	0.2152	0
100	Comedy	Crime	4.132	4.226	0.2316	0
181	Comedy	Action	3.978	4.065	0.2152	0
258	Comedy	Drama	3.768	3.843	0.2978	0

```
[12:44:00] INFO     Feature catalog exported to reports/feature_catalog.md (26 features)
```

Stage 6 — Model Training & MLflow Run Logging (`python3 -m src.training.train`)

Stage 6: Model Training & MLflow Tracking — zsh · Python 3.11

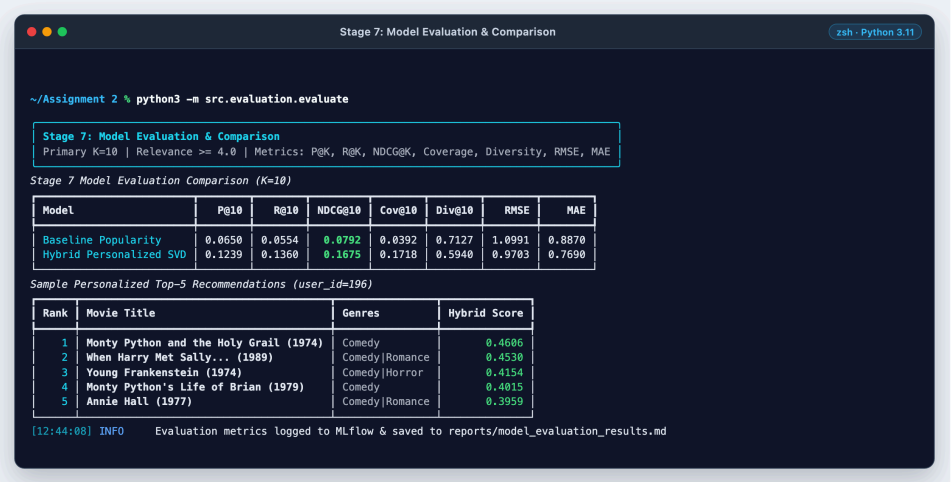
```
~/Assignment 2 % python3 -m src.training.train

┌──────────────────────────────────────────────────────────┐
│ Stage 6: Model Training & MLflow Tracking                │
│ Experiment: movielens-100k-recommender | URI: sqlite:///models/mlflow.db │
└──────────────────────────────────────────────────────────┘
2026/11/11 12:44:03 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/11/11 12:44:03 INFO mlflow.store.db.utils: Updating database tables
2026/11/11 12:44:03 INFO mlflow.tracking.fluent: Experiment with name 'movielens-100k-recommender' does not exist. Creating a new experiment.
Stage 6 Trained Models & MLflow Runs
```

Model	Type	Train Time	Train RMSE	MLflow Run ID
PopularityRecommender	Baseline	0.021s	–	9f99d033cce8...
HybridSVDRecommender	Personalized Hybrid	1.195s	0.8194	8ee6abc64cd2...

```
[12:44:05] INFO     Models & MLflow runs logged; registry saved to models/model_registry.json
```

Stage 7 — Model Evaluation & Comparison (`python3 -m src.evaluation.evaluate`)



Stage 7: Model Evaluation & Comparison — zsh · Python 3.11

```
~/Assignment 2 % python3 -m src.evaluation.evaluate

Stage 7: Model Evaluation & Comparison
Primary K=10 | Relevance >= 4.0 | Metrics: P@K, R@K, NDCG@K, Coverage, Diversity, RMSE, MAE

Stage 7 Model Evaluation Comparison (K=10)
```

Model	P@10	R@10	NDCG@10	Cov@10	Div@10	RMSE	MAE
Baseline Popularity	0.0650	0.0554	0.0792	0.0392	0.7127	1.0991	0.8870
Hybrid Personalized SVD	0.1239	0.1360	0.1675	0.1718	0.5940	0.9703	0.7690

Sample Personalized Top-5 Recommendations (user_id=196)

Rank	Movie Title	Genres	Hybrid Score
1	Monty Python and the Holy Grail (1974)	Comedy	0.4606
2	When Harry Met Sally... (1989)	Comedy|Romance	0.4530
3	Young Frankenstein (1974)	Comedy|Horror	0.4154
4	Monty Python's Life of Brian (1979)	Comedy	0.4015
5	Annie Hall (1977)	Comedy|Romance	0.3959

```
[12:44:08] INFO     Evaluation metrics logged to MLflow & saved to reports/model_evaluation_results.md
```

Stage 8 — Prefect 3 End-to-End Orchestration Flow Run (`python3 -m dags.recommendation_pipeline_dag`)



Stage 8: End-to-End Prefect Orchestration Run — zsh · Python 3.11

```
~/Assignment 2 % python3 -m dags.recommendation_pipeline_dag
12:44:12.647 | INFO    | prefect - Starting temporary server on http://127.0.0.1:8052
See https://docs.prefect.io/v3/concepts/server#how-to-guides for more information on running a dedicated Prefect server.
12:44:14.855 | INFO    | Flow run 'literate-ladybug' - Beginning flow run 'literate-ladybug' for flow 'recommendation-system-data-pipeline'

Stage 8: End-to-End Prefect Orchestration Flow
Flow: recommendation-system-data-pipeline | Cron: 0 2 * * *

Stage 1: Data Ingestion
Source: ml-100k | Staging Dir: data/raw
... [intermediate stages 1-6 executed cleanly; see Stages 1-6 screenshots] ...

Stage 7: Model Evaluation & Comparison
Primary K=10 | Relevance >= 4.0 | Metrics: P@K, R@K, NDCG@K, Coverage, Diversity, RMSE, MAE

Stage 7 Model Evaluation Comparison (K=10)
```

Model	P@10	R@10	NDCG@10	Cov@10	Div@10	RMSE	MAE
Baseline Popularity	0.0650	0.0554	0.0792	0.0392	0.7127	1.0991	0.8870
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4	Monty Python's Life of Brian (1979)	Comedy	0.4015
5	Annie Hall (1977)	Comedy|Romance	0.3959

```
[12:44:19] INFO     Evaluation metrics logged to MLflow & saved to reports/model_evaluation_results.md
12:44:19.356 | INFO    | Task run 'stage-7-model-evaluation-58d' - Finished in state Completed()
Stage 8 End-to-End Orchestration Run Summary
```

DAG Stage	Task Status	Duration (s)
1. Data Ingestion	COMPLETED	0.468s
2. Data Validation & Quarantine	COMPLETED	0.340s
3. Data Preparation & Temporal Split	COMPLETED	0.483s
4. Data Transformation & Matrices	COMPLETED	0.183s
5. Feature Management & Catalog	COMPLETED	0.119s
6. Model Training & MLflow Tracking	COMPLETED	2.403s
7. Model Evaluation & Reporting	COMPLETED	0.471s
Total Pipeline	SUCCESS	4.497s

```
[12:44:19] INFO     Orchestration summary saved to reports/orchestration_run_summary.json
12:44:20.043 | INFO    | Flow run 'literate-ladybug' - Finished in state Completed()
12:44:20.054 | INFO    | prefect - Stopping temporary server on http://127.0.0.1:8052
```

MLflow 3.18 Experiment Tracking UI — Logged Runs (`movielens-100k-recommender`)

